

Joy Roy, PhD

CONTACT

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SKILLS

Languages
Python, SQL, R, Bash, C/C++

AI / ML
Deep Learning, NLP, LLMs,
Agent Systems, RAG,
PyTorch, Scikit-learn

Data & Imaging
Pandas, NumPy, ETL & ML
pipelines, multi-site data
harmonization, MRI/fMRI

Cloud & Systems
Microsoft Azure, GCP,
Docker, Linux/Unix, Git,
HPC, Slurm

Clinical & Regulatory
Clinical AI validation,
IRB development,
study design, scientific writing,
clinical usability studies

EDUCATION

Ph.D. Biomedical Informatics
University of Pittsburgh
School of Medicine
2020 – 2024

B.S. Bioinformatics & Computational Biology
2015 – 2019
University of Maryland
Baltimore County

B.A. Mathematics
2015 – 2019
University of Maryland
Baltimore County

WORK EXPERIENCE

Senior Data Scientist Sept 2025 – Present
Mayo Clinic Remote – Baltimore, MD

- Lead data scientist for AI-enabled Tumor Board initiatives, directing clinical needs assessment, pilot development, and evaluation across multiple oncology use cases.
- Provide data science consulting across Mayo Clinic Comprehensive Cancer Center AI projects spanning multiple disease groups and imaging modalities, including WSI, CT, and MRI.
- Investigate and evaluate medical imaging foundation models and emerging methods for representation learning, segmentation, and cancer prognosis.
- Directed evaluation of Microsoft Healthcare Agent Orchestrator (HAO), establishing quantitative and clinician-informed frameworks for clinical utility, safety, factuality, and workflow impact.
- Partner with physicians, engineers, and external collaborators to translate clinical problems into technically feasible AI solutions and implementation strategies.
- Support research and regulatory efforts by writing 4 IRB protocols and serving as lead author on 1 manuscript and 2 conference abstracts.

Health Informatics Scientist Jan 2025 - Sept 2025
Henry M Jackson Foundation Bethesda, MD

- Designed a framework to harmonize heterogeneous research datasets into a centralized data lake, enabling downstream machine learning and analytics.
- Led cross-disciplinary discussions with physicians, psychologists, and researchers to define requirements for data storage, integration, and longitudinal tracking.
- Consulted on software and hardware requirements to support deployment of research and analytics infrastructure.
- Appointed to Walter Reed Hospital with Interim DOD Secret level Clearance.

PhD Research Assistant 2020 – 2024
Children’s Hospital of Pittsburgh Pittsburgh, PA

- Developed open-source command-line software using Bash, Python, and Docker.
- Developed, trained, and evaluated machine learning models (e.g., Random Forests, CNNs) to identify imaging-based predictors of clinical outcomes from large-scale medical datasets.
- Designed, implemented, and deployed MRI preprocessing pipelines to enable data-driven analysis while ensuring data quality and clinical validity.
- Trained deep learning models on Google Cloud Compute Engine and HPC clusters.

Software Engineer 2019 – 2020
Hughes Network Systems Gaithersburg, MD

- Developed real-time C/C++ software for satellite communication, modems, and network protocols.
- Maintained Jenkins pipelines for automated unit testing and CI/CD.

Bioinformatics Research Assistant 2017 – 2019
Univ. of Maryland Baltimore County Baltimore, MD

- Led development of C/C++ software for biological data curation, contributing to a peer-reviewed publication (*Bioinformatics*, 2020).

PUBLICATIONS

- **Roy J** et al. A Retrospective Evaluation of the Microsoft Healthcare Agent Orchestrator for Tumor Board Patient Summaries *medrXiv*. 2026.
- **Roy J** et al. Functional network organization is locally atypical in children with congenital heart disease. *NeuroImage: Clinical*. 2026.

Full list: orcid.org/0000-0002-0931-7253